

Date: _____

(1) What does the CPU divide a memory address in to Tag, Index, and offset when accessing a cache, instead of sending the full address directly to compare against all cache entries?

→ The CPU divides a memory address into: Tag, Index and offset because searching the entire cache using the full address would be slow and expensive.

- Index → selects the cache line/set quickly
- Tag → checks if the correct block is present.
- Offset → selects the exact byte/word inside the block

This makes cache access faster and more efficient.

(2) Explain in detail the three main cache mapping techniques:

(1) Direct Mapping

- Each memory block maps to one fixed cache line
$$\text{cache line} = (\text{Block Number}) \bmod (\text{Number of lines})$$

Pros

- Simple
- Fast
- Cheap

Cons

- High conflict misses

(2) Fully Associative Mapping

- A memory block can be placed in any cache line.

Pros

- Very few conflict misses
- Best cache utilization

Cons

- Expensive hardware
- Slower due to searching all lines

(3) Set-Associative Mapping

- Cache is divided in to sets.
- A block maps to one set but can go into any line inside that set.

$$\text{Set Number} = (\text{Block Number}) \bmod (\text{Number of sets})$$

Pros

- Balanced performance
- Fewer conflicts than direct mapping

Cons

- More complex than direct mapping.